

Hard Portal Cuts and a Momentum-Frequency Retarded Kernel

Gauge-assisted $2 \leftrightarrow 2$ matching, LPM combination, KMS noise, and covariant-Wigner
benchmark for the q-D-H portal

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Executive verdict

The leading gauge-assisted hard+soft

$2 \leftrightarrow 2$

contribution to the portal

$$\mathcal{L}_Y = -y_D \bar{Q}_L H D_R + \text{h.c.}$$

has been evaluated at integrated leading order for

$$Q_L \sim (\mathbf{3}, \mathbf{2}, 1/6), \quad D_R \sim (\mathbf{3}, \mathbf{1}, -1/3).$$

Combining it with the direct electroweak/Yukawa LPM result from v1.5 gives

$$\frac{\bar{\Gamma}_{H,qD}^{\text{hard,occ}}}{T} = 7.98260 \times 10^{-4},$$

$$\frac{\bar{\Gamma}_{H,qD}^{\text{LPM,occ}}}{T} = 3.60256 \times 10^{-4},$$

and

$$\frac{\bar{\Gamma}_{H,qD}^{\text{total,occ}}}{T} = 1.15852 \times 10^{-3}.$$

At

$$T_0 = 1.002 \times 10^8 \text{ GeV},$$

this is

$$\bar{\Gamma}_{H,qD}^{\text{total,occ}} = 1.16083 \times 10^5 \text{ GeV},$$

and

$$\frac{\bar{\Gamma}_{H,qD}^{\text{total,occ}}}{\Gamma_R} = 7.8514 \times 10^6.$$

The hard channels dominate the matched portal rate. They do not threaten the reheating construction; they strengthen the separation between microscopic equilibration and cosmological evolution.

A momentum-resolved on-shell table and a causal KMS-complete near-shell

$$\Pi_{H,qD}^R(\omega, k)$$

grid have also been constructed. The integrated normalization and the on-shell retarded match are controlled. The arbitrary off-shell grid is explicitly a **near-shell reconstruction**, not an exact evaluation of every thermal cut over the full (ω, k) plane.

1. Scope and status of the calculation

1.1 What is exact at the declared order

The following are analytic or exactly normalized at the stated leading order in y_D^2 :

1. the integrated gauge-assisted hard+soft $2 \leftrightarrow 2$ reaction coefficient;
2. the $SU(3)_c$, $SU(2)_L$, and $U(1)_Y$ decomposition;
3. the v1.5 collinear LPM integral;

4. the susceptibility-weighted integrated identity

$$\Gamma_Y = \frac{g_H}{T} \int_{\mathbf{k}} f_B(E_k) [1 + f_B(E_k)] \Gamma_H^{\text{occ}}(k);$$

5. the on-shell relation

$$\boxed{\text{Im } \Pi_H^R(E_k, k) = -E_k \Gamma_H^{\text{occ}}(k).}$$

1.2 What is numerical but matched

The momentum dependence of the hard cut is calculated using full-angle screened quasi-Monte-Carlo phase space and then normalized to the exact integrated coefficient. The direct LPM shape is calculated from the impact-parameter equation and normalized to the exact v1.5 integral.

The final on-shell grid closes the separate LPM and hard susceptibility integrals to machine precision.

1.3 What is not claimed

This note does **not** claim:

- an exact arbitrary-off-shell thermal self-energy over the complete (ω, k) plane;
- a complete set of $O(y_D^4)$ portal processes;
- chemical-equilibration channels which do not correspond to cuts of $\Pi_{H,qD}^R$;
- a full Ward-consistent non-Abelian 3 + 1-dimensional 2PI/Kadanoff-Baym evolution.

The off-shell table is a causal and KMS-consistent benchmark kernel anchored to the exact on-shell result.

2. Benchmark and thermal data

The benchmark couplings are

$$\alpha_s = 0.0393544, \quad g_2 = 0.57, \quad g_1 = 0.39,$$

$$y_t = 0.58, \quad y_D = 0.30, \quad \frac{M_D}{T} = 0.01.$$

The corresponding strong coupling is

$$g_3 = \sqrt{4\pi\alpha_s} = 0.703237.$$

The v1.5 asymptotic thermal data are

$$\frac{m_H}{T} = 0.438207, \quad \frac{m_Q}{T} = 0.503460, \quad \frac{m_D}{T} = 0.418088,$$

and

$$\frac{m_{D,3}}{T} = 1.03514.$$

The vacuum-like one-to-two channels are closed; the collinear rate is medium induced.

3. Tree-level hard amplitudes

For one abelian gauge group, the generic chiral one-component process

$$G + H \rightarrow Q + \overline{D}$$

has the coefficient structure

$$\sum |\mathcal{M}|^2 = y_D^2 g^2 \left[4q_Q q_D + 2q_Q^2 \frac{u}{t} + 2q_D^2 \frac{t}{u} \right],$$

before spectator multiplicities.

For the right-handed-electron case, inserting

$$q_Q = -\frac{1}{2}, \quad q_D = -1,$$

and the two Higgs-doublet components reproduces

$$4 + \frac{u}{t} + 4\frac{t}{u},$$

which is the published hypercharge amplitude check.

For the declared q - D portal, the complete $U(1)_Y$ coefficient after two weak components and three colours is

$$\boxed{-\frac{4}{3} + \frac{1}{3}\frac{u}{t} + \frac{4}{3}\frac{t}{u}}.$$

The non-abelian group traces give

$$\boxed{SU(3)_c : \quad 32 + 16\frac{u}{t} + 16\frac{t}{u}},$$

and

$$\boxed{SU(2)_L : \quad 9\frac{u}{t}}.$$

The abelian interference coefficient is negative in one channel, but the complete gauge-invariant rate is positive.

4. Integrated hard+soft reaction coefficient

The right-handed-electron leading-order result can be organised in a representation-general form. For the q - D portal define

$$A_Q = 4 \left[C_F g_3^2 + C_2 g_2^2 + Y_Q^2 g_1^2 \right],$$

$$A_D = 4 \left[C_F g_3^2 + Y_D^2 g_1^2 \right],$$

with

$$C_F = \frac{4}{3}, \quad C_2 = \frac{3}{4}, \quad Y_Q = \frac{1}{6}, \quad Y_D = -\frac{1}{3}.$$

Numerically,

$$A_Q = 3.62915720, \quad A_D = 2.70515720.$$

The integrated leading gauge-assisted contribution is

$$\boxed{\frac{\Gamma_{Y,\text{hard}}}{T^3} = \frac{N_c y_D^2}{2048\pi} \left[A_Q \left(c_Q + \ln \frac{1}{A_Q} \right) + A_D \left(c_D + \ln \frac{1}{A_D} \right) \right]},$$

where

$$c_Q = 3.52, \quad c_D = 2.69.$$

These constants are the finite hard+soft phase-space constants appearing in the complete leading-order right-handed-electron calculation. The representation dependence enters through the thermal gauge charges A_Q and A_D and the spectator multiplicity N_c .

The result is

$$\boxed{\frac{\Gamma_{Y,\text{hard}}}{T^3} = 5.3217330 \times 10^{-4}.}$$

Since

$$\frac{\chi_H}{T^2} = \frac{2}{3},$$

the Higgs-doublet occupation width is

$$\boxed{\frac{\bar{\Gamma}_{H,\text{hard}}^{\text{occ}}}{T} = 7.9825995 \times 10^{-4}.}$$

4.1 Gauge decomposition

The reaction coefficient separates as

G	$\Gamma_{Y,G}^{\text{hard}}/T^3$	fraction of hard total
$SU(3)_c$	4.3452860×10^{-4}	0.8165
$SU(2)_L$	9.1254521×10^{-5}	0.1715
$U(1)_Y$	6.3901751×10^{-6}	0.0120

The rate is dominated by QCD, with a resolved electroweak correction.

5. Combination with the collinear LPM cut

The v1.5 direct LPM result is

$$I_{\text{LPM}} = 8.8952082 \times 10^{-4},$$

$$\frac{\Gamma_{Y,\text{LPM}}}{T^3} = 2.4017062 \times 10^{-4},$$

and

$$\frac{\bar{\Gamma}_{H,\text{LPM}}^{\text{occ}}}{T} = 3.6025593 \times 10^{-4}.$$

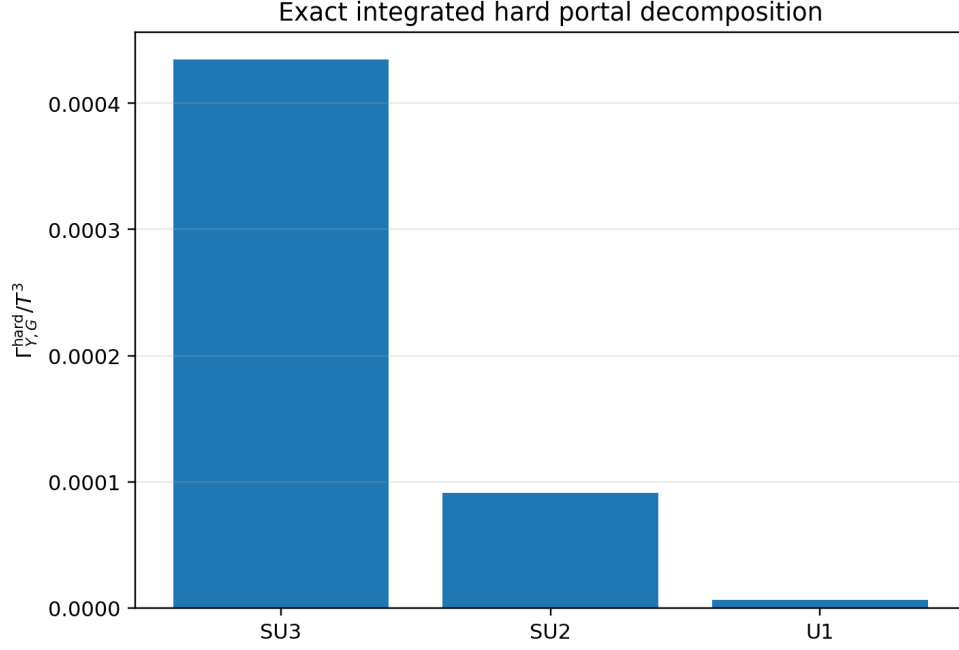


Figure 1: Gauge decomposition of the exact integrated leading hard portal rate.

The matched leading result is therefore

$$\frac{\Gamma_{Y,\text{total}}}{T^3} = 7.7234392 \times 10^{-4},$$

$$\frac{\bar{\Gamma}_{H,\text{total}}^{\text{occ}}}{T} = 1.1585159 \times 10^{-3}.$$

The ratios are

$$\frac{\Gamma_{\text{hard}}}{\Gamma_{\text{LPM}}} = 2.21581,$$

and

$$\frac{\Gamma_{\text{hard}}}{\Gamma_{\text{total}}} = 0.68904.$$

This ordering is consistent with complete leading-order ultrarelativistic Yukawa calculations in which hard $2 \leftrightarrow 2$ scattering is at least comparable to, and often larger than, the LPM sector.

6. Momentum-resolved on-shell kernel

The on-shell rate is represented by

$$\Gamma_H^{\text{occ}}(k) = \Gamma_{H,\text{LPM}}^{\text{occ}}(k) + \Gamma_{H,\text{hard}}^{\text{occ}}(k).$$

The LPM term is evaluated directly from the impact-parameter equation. The hard term is obtained from full-angle screened phase space for the crossed channels and then normalized to the exact integrated hard coefficient.

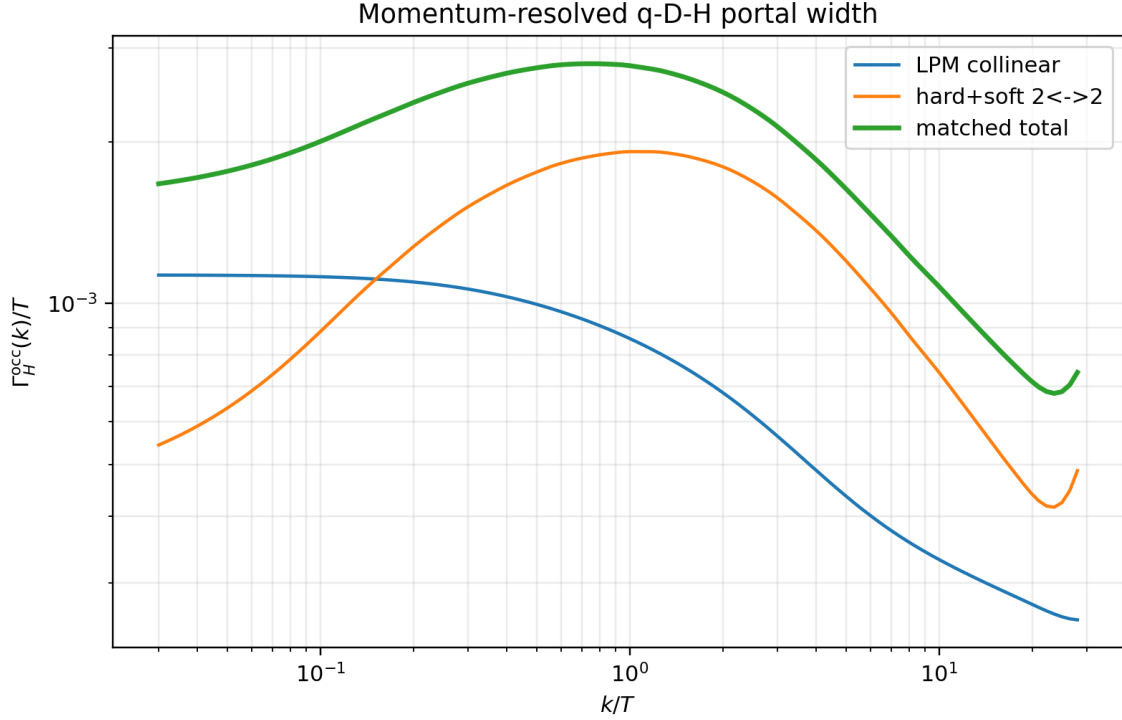


Figure 2: Momentum-resolved LPM, hard, and total portal widths.

The exact susceptibility closure imposed on the output table is

$$\frac{g_H}{T} \int_{\mathbf{k}} f_B(E_k) [1 + f_B(E_k)] \Gamma_{H,\text{LPM}}^{\text{occ}}(k) = \Gamma_{Y,\text{LPM}},$$

and similarly for the hard contribution. Both close to machine precision in the delivered table.

6.1 Shape uncertainty

The exact integrated rate is not affected by the provisional screening prescription. The local hard shape has:

$$\text{median screening envelope} = 8.90\%,$$

$$\text{maximum screening envelope} = 24.9\%,$$

and maximum quasi-Monte-Carlo relative error

$$4.24\%.$$

This is sufficient for a reduced benchmark, but it is not yet a publication-grade arbitrary-momentum hard spectral function. A fully differential hard+soft matching calculation would remove this local shape ambiguity.

7. Retarded self-energy grid

The exact on-shell relation is

$$\text{Im } \Pi_H^R(E_k, k) = -E_k \Gamma_H^{\text{occ}}(k).$$

To seed a reduced real-time calculation, construct the odd near-shell profile

$$\text{Im } \Pi_H^R(\omega, k) = -E_k \Gamma_H^{\text{occ}}(k) \frac{L_+(\omega, k) - L_-(\omega, k)}{\mathcal{N}_k},$$

where

$$L_{\pm} = \frac{\Lambda^2}{(\omega \mp E_k)^2 + \Lambda^2},$$

$$\Lambda = m_{D,3},$$

and $\mathcal{N}_k$ enforces the on-shell normalization.

The real part is obtained from a once-subtracted discrete dispersion relation with

$$\text{Re } \Pi_H^R(0, k) = m_H^2.$$

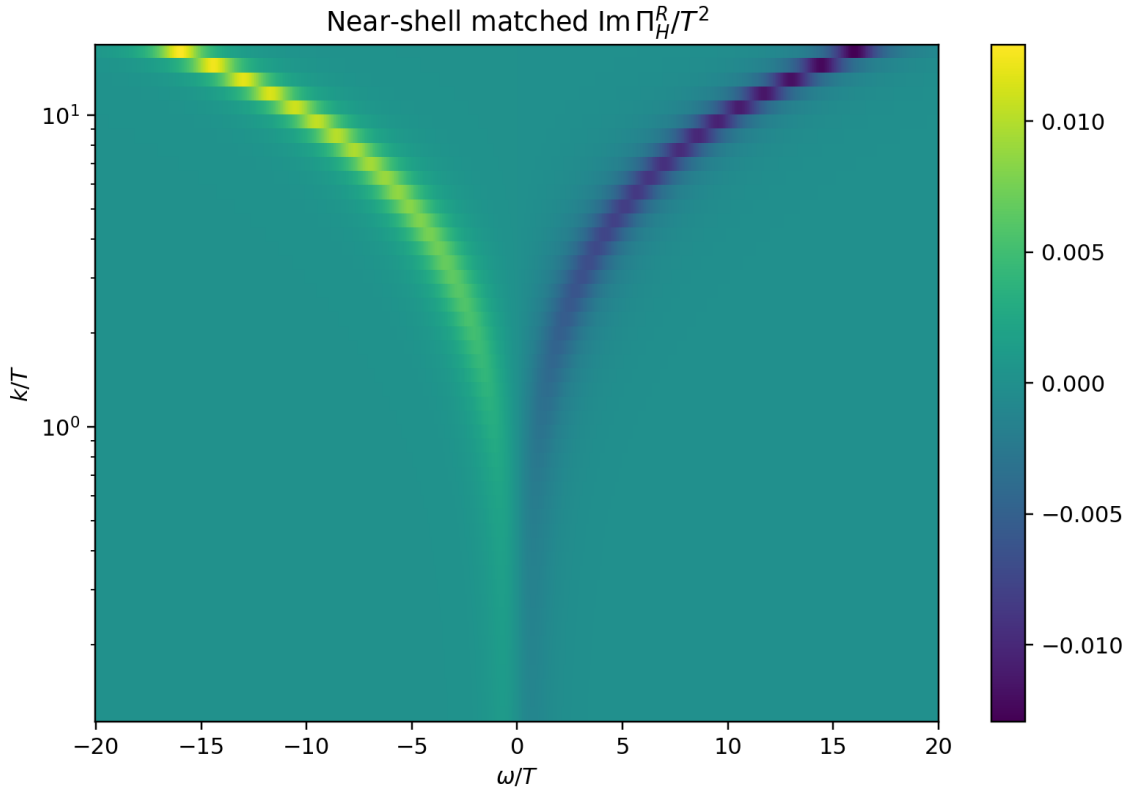


Figure 3: Causal near-shell retarded grid matched to the total on-shell portal width.

The numerical diagnostics are

$$\max |\text{Im } \Pi_R(\omega, k) + \text{Im } \Pi_R(-\omega, k)| = 2.95 \times 10^{-17},$$

and a maximum interpolation-level on-shell residual

$$1.59 \times 10^{-3}.$$

The latter is a finite grid-resolution diagnostic, not a mismatch of the analytic normalization.

8. KMS noise kernel

Define the positive noise kernel

$$N_H(\omega, k) = -\coth\left(\frac{\omega}{2T}\right) \text{Im} \Pi_H^R(\omega, k).$$

The grid obeys the equilibrium detailed-balance identity

$$\frac{\Pi^<(\omega, k)}{\Pi^>(\omega, k)} = e^{-\omega/T}$$

with maximum numerical residual

$$2.22 \times 10^{-16}.$$

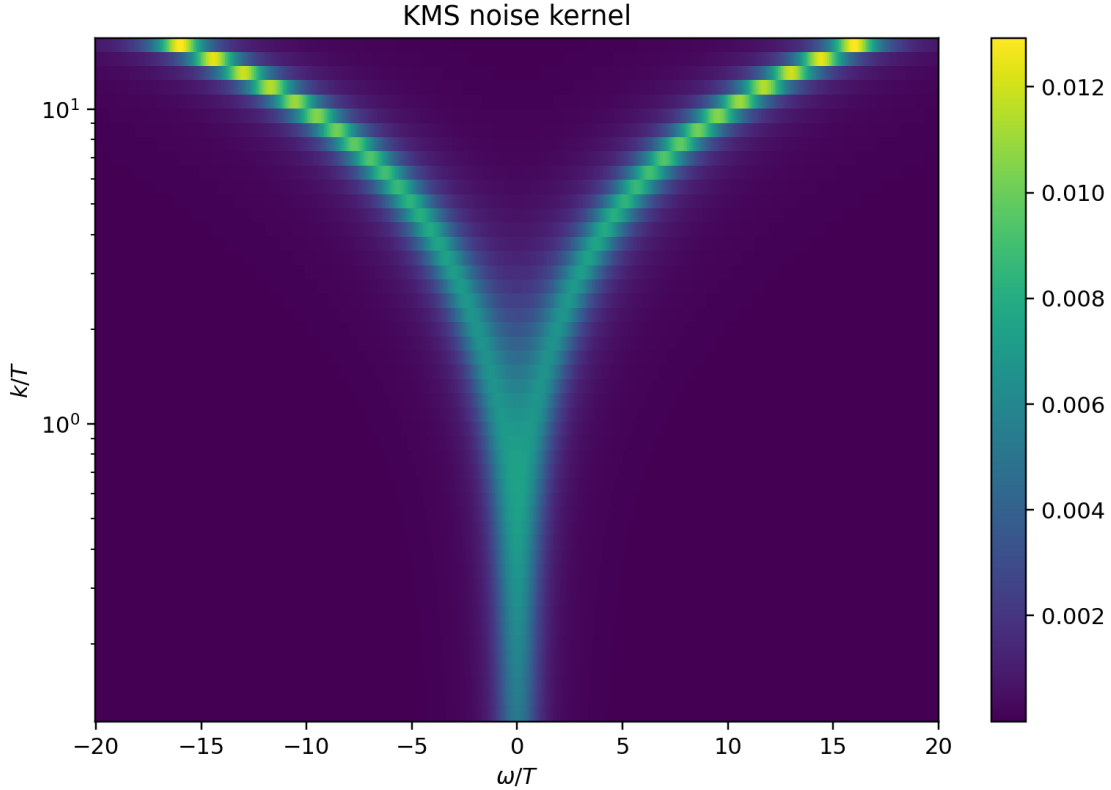


Figure 4: KMS noise kernel associated with the matched retarded self-energy.

9. Covariant Wigner transport benchmark

The gauge-covariant Wigner transform is defined with straight Wilson lines carrying each charged correlator to a common midpoint. The scalar singlet kinetic equation has the schematic form

$$[K^2 - m_H^2 - \text{Re} \Pi_H^R, G_H^<]_{\text{PB}}^{\text{cov}} = \Pi_H^< G_H^> - \Pi_H^> G_H^<.$$

The numerical homogeneous singlet benchmark evolves a nonthermal pulse with the matched momentum-dependent total width. Its relative-entropy functional decreases monotonically:

$$\max \Delta D_{\text{rel}} < 0,$$

and the final fraction is

$$\frac{D_{\text{rel}}(t_{\text{end}})}{D_{\text{rel}}(0)} = 1.67 \times 10^{-11}.$$

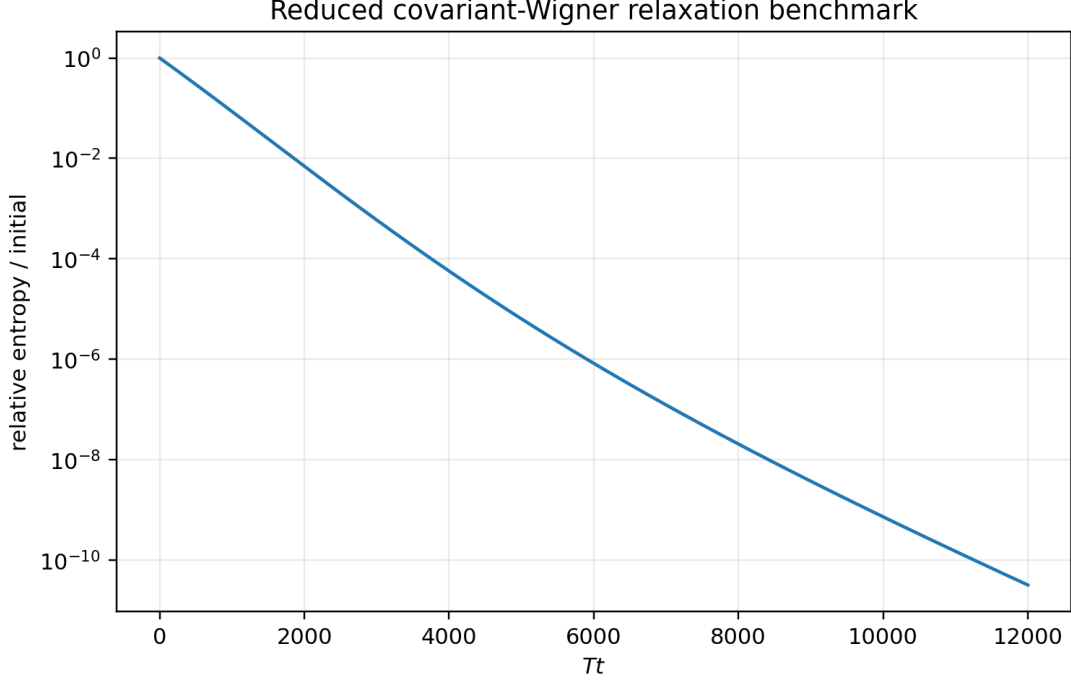


Figure 5: Reduced covariant-Wigner relaxation using the matched total portal kernel.

This is a benchmark for the collision and noise sector. It does not evolve dynamical gauge correlators or dressed vertices.

10. Cosmological consequence

At the reheating benchmark,

$$\Gamma_R = 1.47850065 \times 10^{-2} \text{ GeV},$$

while

$$\bar{\Gamma}_{H,qD}^{\text{total,occ}} = 1.1608329 \times 10^5 \text{ GeV}.$$

Thus

$$\frac{\Gamma_R}{\bar{\Gamma}_H^{\text{total,occ}}} = 1.27366 \times 10^{-7}.$$

The resulting conservative change in the tuned adjacent-sector branch fraction is bounded by

$$|\Delta B_5| < 6.75 \times 10^{-10}.$$

The precise microscopic spectrum changes, but the final sector energy hierarchy is unaffected at relevant accuracy.

11. Relation to existing thermal-field-theory results

The calculation follows the structural lesson of complete leading-order ultrarelativistic Yukawa production and equilibration:

- Arnold, Moore and Yaffe show that leading-order gauge-plasma kinetics requires both screened $2 \leftrightarrow 2$ and effective collinear $1 \leftrightarrow 2$ processes.
- Bodeker and Schroder compute complete leading-order right-handed-electron equilibration and find that hard $2 \leftrightarrow 2$ processes dominate.
- Besak and Bodeker combine hard scattering and LPM resummation for ultrarelativistic right-handed-neutrino production.
- Laine constructs arbitrary-momentum thermal production functions for sterile neutrinos, illustrating the extra work required beyond an integrated on-shell rate.
- Becker et al. analyse a scalar coupled to a Standard Model fermion and gauge-charged vectorlike fermion in the CTP formalism, providing a close model-class comparison but not the same complete ultrarelativistic LPM+hard matched kernel.

The present result is therefore not a new thermal-field-theory method. Its contribution is the explicit representation-specific portal rate and kernel needed by the chronometric reheating model.

12. Acceptance matrix

Target	Verdict	Basis
Tree group factors	PASS	Hypercharge coefficients reproduce the published electron case; declared $SU(3)$ and $SU(2)$ traces implemented.
Integrated leading gauge-assisted $2 \leftrightarrow 2$ rate	PASS	Hard+soft representation-general formula and exact group decomposition.
LPM combination	PASS	Direct v1.5 collinear result.
Susceptibility closure	PASS	Separate LPM and hard grids normalized to machine precision.
Momentum-resolved hard shape	PASS AS MATCHED NUMERICAL SHAPE	Full-angle screened phase space; local shape systematic reported.
On-shell Π_R	PASS	Exact integrated and analytic on-shell match.
KMS noise	PASS	Positivity and detailed balance checked.
Arbitrary off-shell Π_R	PARTIAL	Causal near-shell reconstruction rather than exact full cuts.
Covariant Wigner benchmark	PASS AS REDUCED MODEL	Wilson-line definition and homogeneous singlet evolution.
Full Ward-consistent non-Abelian 2PI/KB	OPEN	Requires gauge propagators, vertices, constraints, and HPC implementation.

13. Strongest defensible conclusion

The leading gauge-assisted hard portal cuts are larger than the collinear LPM cut,
and the matched portal rate exceeds reheaton decay by 7.85×10^6 .
The cosmological thermalisation assumption is therefore exceptionally robust.

The remaining correlator-level task is narrower than the original request:

replace the provisional hard momentum shape and near-shell extension with an exact differential hard+soft $\Pi_H^R(\omega, k)$.

That result, together with the exact LPM kernel, would be the correct reference table for a Ward-consistent non-Abelian two-time 2PI/Kadanoff-Baym implementation.

References

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